

1. 1.1 Evaluate the expression of $\frac{(-\sqrt{3} + 3i)(1-i)}{(3 + \sqrt{3}i)i(\sqrt{3} + \sqrt{3}i)}$. (5 marks)

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1.2 Given two complex numbers $z_1 = \frac{4i^{44}}{1-i^{2020}}$ and $z_2 = \frac{3i^{2560} + 21i^{31}}{5i}$. Compute the expression of $z_1^2 \cdot z_2^3$. (5 marks)

2. 2.1 If the complex number z satisfies both the conditions

$$\operatorname{Im}(i + \bar{z}) = 4 \text{ and } \left| \frac{z-4}{z-8} \right| = 1.$$

Then, evaluating the value of $z\bar{z}$. (5 marks)

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2.2 Find all roots of the complex equation. (5 marks)

$$z^3 + \frac{\sqrt{3}}{2} + \frac{i}{2} = 0$$

3. Find all roots of the complex equation $z^4 \bar{z} + 243 = 0$. (10 marks)

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4. Let $P(z) = z^8 - \frac{5}{2}z^4 + 1$. The complex number w is a root of $P(z) = 0$. (10 marks)

4.1 Show that iw and $\frac{1}{w}$ are also roots of $P(z) = 0$. (5 marks)

4.2 Find all roots of $P(z) = 0$. (5 marks)

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5. Let $A = \begin{bmatrix} 3 & -1 & b-1 \\ a+b & 0 & 2 \\ c & -4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2a & 6 \\ -1 & 0 & -4 \\ \frac{c}{2} & 2 & 1 \end{bmatrix}$ and $A = B^T$

Find the symmetric matrix (S) and skew-symmetric matrix (K) of $2A$. (10 marks)

6. Let $A^3 \cdot \begin{bmatrix} 1 & 4 & 3 \\ -2 & 1 & -5 \\ 1 & -1 & 2 \end{bmatrix} = (-2) \cdot \begin{bmatrix} 0 & 2 & -3 \\ -2 & 1 & 2 \\ 1 & -1 & 2 \end{bmatrix}^T$. Evaluate $\det(A)$. (10 marks)

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7. If $\begin{bmatrix} 2 & 1 \\ 4 & a \end{bmatrix}$ is a singular matrix and $\begin{bmatrix} 1 & a & -a \\ 0 & -1 & 3 \\ -a & 0 & 0 \end{bmatrix} \cdot A^{-1} = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 3 \\ 1 & 0 & 1 \end{bmatrix}$.

Find the value of the constant a and compute the matrix A . (10 marks)

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8 8.1 Solve the system of equations using Gauss–Jordan elimination (5 marks)

$$\begin{aligned}x_1 - x_2 + 2x_3 - x_4 &= -1, \\2x_1 + x_2 - 2x_3 - x_4 &= -3, \\-x_1 + 2x_2 - 4x_3 + x_4 &= 1, \\3x_1 &\quad - 2x_4 = -4.\end{aligned}$$

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8.2 Solve the system of three equations of three variables (5 marks)

$$2x + y - 2z = 6,$$

$$5x + 2y + 1 = 0,$$

$$-4x - 5y + 8z = 6.$$

By Cramer's rule and compute the value of (x, y, z) .

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